

Lab 7

q1

```
1: bless@cis106: ~/Documents/Csv ▾
bless@cis106:~/Documents$ cd Csv/
bless@cis106:~/Documents/Csv$ ls
cars.csv  cereal.csv  contacts.csv  country.csv  film.csv
bless@cis106:~/Documents/Csv$ cat cereal.csv
name;mfr;type;calories;protein;fat;sodium;fiber;carbo;sugars;potass;vitamins;shelf;weight;cups;rating
String;Categorical;Categorical;Int;Int;Int;Int;Int;Float;Float;Int;Int;Int;Int;Float;Float;Float
100% Bran;N;C;70;4;1;130;10;5;6;280;25;3;1;0.33;68.402973
100% Natural Bran;Q;C;120;3;5;15;2;8;8;135;0;3;1;1;33.983679
All-Bran;K;C;70;4;1;260;9;7;5;320;25;3;1;0.33;59.425505
All-Bran with Extra Fiber;K;C;50;4;0;140;14;8;0;330;25;3;1;0.5;93.704912
Almond Delight;D;C;110;2;2;200;1;14;8;1;25;2;1;0.75;34.284842

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String;Categorical;Categorical;Int;Int;Int;Int;Int;Float;Float;Int;Int;Int;Int;Float;Float;Float
100% Bran;N;C;70;4;1;130;10;5;6;280;25;3;1;0.33;68.402973
100% Natural Bran;Q;C;120;3;5;15;2;8;8;135;0;3;1;1;33.983679
All-Bran;K;C;70;4;1;260;9;7;5;320;25;3;1;0.33;59.425505
All-Bran with Extra Fiber;K;C;50;4;0;140;14;8;0;330;25;3;1;0.5;93.704912
Almond Delight;D;C;110;2;2;200;1;14;8;1;25;2;1;0.75;34.284842

bless@cis106:~/Documents/Csv$ cat -nEv cereal.csv
 1 name;mfr;type;calories;protein;fat;sodium;fiber;carbo;sugars;potass;vitamins;shelf;weight;cups;rating^M$
 2 String;Categorical;Categorical;Int;Int;Int;Int;Int;Float;Float;Int;Int;Int;Int;Float;Float;Float^M$
 3 100% Bran;N;C;70;4;1;130;10;5;6;280;25;3;1;0.33;68.402973^M$

bless@cis106:~/Documents/Csv$ head -n 2 cereal.csv
name;mfr;type;calories;protein;fat;sodium;fiber;carbo;sugars;potass;vitamins;shelf;weight;cups;rating
String;Categorical;Categorical;Int;Int;Int;Int;Int;Float;Float;Int;Int;Int;Int;Float;Float;Float
bless@cis106:~/Documents/Csv$ tail -n 2 cereal.csv
Wheaties;G;C;100;3;1;200;3;17;3;110;25;1;1;1;51.592193
Wheaties Honey Gold;G;C;110;2;1;200;1;16;8;60;25;1;1;0.75;36.187559
bless@cis106:~/Documents/Csv$
```

q2

```
1: bless@cis106: ~/Documents/Csv ▾
bless@cis106:~/Documents/Csv$ head -n 1 cars.csv
Car;MPG;Cylinders;Displacement;Horsepower;Weight;Acceleration;Model;Origin
bless@cis106:~/Documents/Csv$ cut -d";" -f1 cars.csv
Car
STRING
Chevrolet Chevelle Malibu

1: bless@cis106: ~/Documents/Csv ▾
bless@cis106:~/Documents/Csv$ cut -d";" -f1,3 cars.csv
Car;Cylinders
STRING;INT
Chevrolet Chevelle Malibu;8
Buick Skylark 320;8
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Chevrolet Camaro = 4
Ford Mustang GL = 4
Volkswagen Pickup = 4
Dodge Rampage = 4
Ford Ranger = 4
Chevy S-10 = 4
bless@cis106:~/Documents/Csv$ cut -d";" -f1,3 cars.csv --output-delimiter=" = "
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```
Volvo 244DL;22.0;4;121.0;98.00;2945.;14.5;75;Europe
Volvo 245;20.0;4;130.0;102.0;3150.;15.7;76;Europe
Volvo 264gl;17.0;6;163.0;125.0;3140.;13.6;78;Europe
Volvo Diesel;30.7;6;145.0;76.00;3160.;19.6;81;Europe
bless@cis106:~/Documents/Csv$ sort cars.csv
```

```
name;mfr;type;calories;protein;fat;sodium;fiber;carbo;sugars;potass;vitamins;shelf;weight;cups;rating
String;Categorical;Categorical;Int;Int;Int;Int;Int;Float;Float;Int;Int;Int;Int;Float;Float;Float
bless@cis106:~/Documents/Csv$ sort -t";" -k4 cereal.csv
```

q3

```
bless@cis106:~/Documents/Books$ grep "War" war-and-peace.txt
The Project Gutenberg eBook of War and Peace, by Leo Tolstoy
Title: War and Peace
the Minister of War."
in to the Minister of War. Five minutes later he returned and bowing
corridor to the cabinet where the Minister of War was at work. The
narrowed disdainfully, he entered the room of the Minister of War with
interested the Minister of War less than any of the other matters he was
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1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the "Right
LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE
OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT
LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.
warranties or the exclusion or limitation of certain types of
bless@cis106:~/Documents/Books$ grep -i "War" war-and-peace.txt
```

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bless@cis106:~/Documents/Books$ grep -ic "War" war-and-peace.txt
1255
bless@cis106:~/Documents/Books$ grep -iwc "War" war-and-peace.txt
293
bless@cis106:~/Documents/Books$ grep -i "^peace" war-and-peace.txt
peace. The old man was beside himself: moved the chair on which he was
peace."
peace." But however much they left her in peace she could not now be
peace so long as a single armed enemy remained on Russian soil and told
peace with you. As for the Swedes—it is their fate to be governed by
peace nor war, neither an advance nor a defensive camp at the Drissa
peace and the abolition of war, and secondly, by the fact that when he
peaceful, and glorious fatherland, I should have proclaimed
peace arising from consciousness of having stifled those personal dreams
peace, he said: There can be no peace, for such is the people's will. He
peaceful refuge where he could recover himself, rest, and think over
bless@cis106:~/Documents/Books$ grep -in "^peace" war-and-peace.txt
5542:peace. The old man was beside himself: moved the chair on which he was
20046:peace."
26471:peace." But however much they left her in peace she could not now be
33900:peace so long as a single armed enemy remained on Russian soil and told
34397:peace with you. As for the Swedes—it is their fate to be governed by
35057:peace nor war, neither an advance nor a defensive camp at the Drissa
36617:peace and the abolition of war, and secondly, by the fact that when he
44768:peaceful, and glorious fatherland, I should have proclaimed
51593:peace arising from consciousness of having stifled those personal dreams
58824:peace, he said: There can be no peace, for such is the people's will. He
59768:peaceful refuge where he could recover himself, rest, and think over
bless@cis106:~/Documents/Books$

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65561:The search for these laws has long been begun and the new methods of
65622:The struggle between the old views and the new was long and stubbornly
65963:The Project Gutenberg Literary Archive Foundation is a non-profit
65971:The Foundation's business office is located at 809 North 1500 West,
65987:The Foundation is committed to complying with the laws regulating
bless@cis106:~/Documents/Books$ grep -nw "^The" war-and-peace.txt

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bless@cis106:~/Documents/Books$ ls
bible.txt      great-expectations.txt  war-and-peace.txt
dracula.txt    pride-and-prejudice.txt
bless@cis106:~/Documents/Books$ grep -Rwi "love" ./
./pride-and-prejudice.txt:by allowance" and "loving with personal love." This distinction applies
./pride-and-prejudice.txt:proper thing to love them. And in the sect--fairly large and yet
./pride-and-prejudice.txt:aversion to actual love. I do not know whether the all-grasping hand of
./pride-and-prejudice.txt:his falling in love at all is not extremely probable. Here again I
./pride-and-prejudice.txt:injured in disposition, and thoroughly in love. As for his being in
bless@cis106:~/Documents/Books$ grep -nR "^l...d" ./
./pride-and-prejudice.txt:1720:liked a red coat myself very well--and, indeed, so I do still at my
./pride-and-prejudice.txt:9474:lived. Not like the wild young men now-a-days, who think of nothing but
./pride-and-prejudice.txt:10016:loved her, but for loving her still well enough to forgive all the
./pride-and-prejudice.txt:13948:liked him but Jane; and even feared that with the others it was a
./dracula.txt:5597:lay down, and when I went over to Bersicker he let me stroke his ears

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```
bless@cis106:~/Documents/Books$ grep -wniR "^d[aeiou]..." ./
./pride-and-prejudice.txt:420:Darcy; nearly all of them would have palpitated and fluttered at the
./pride-and-prejudice.txt:1022:dance."
./pride-and-prejudice.txt:1036:Darcy, looking at the eldest Miss Bennet.
./pride-and-prejudice.txt:1110:dear, to have given him one of your set-downs. I quite detest the man."
```

```
bless@cis106:~/Documents/Books$ grep -iE "a{2}" dracula.txt
help me deck the room with my garlic, which is all the way from Haarlem,
Haarlem. To-night Dr. Van Helsing is going away, as he has to be for a
bless@cis106:~/Documents/Books$
```